# Renfei Zhang

## Education

**Carnegie Mellon University** – Pittsburgh, USA — Sep. 2026 - Sep. 2030 (Expected)
*Ph.D. in Societal Computing*
*Advisor: Prof. Niloofar Mireshghallah*

**Simon Fraser University** – Burnaby, Canada — Sep. 2022 - Jun. 2025
*Bachelor of Science in Computing Science*

**McGill University** – Montreal, Canada — Sep. 2015 - Jun. 2022
*Bachelor of Science in Physiology*

## Research Interest

My research focuses on **LLM reasoning** and **alignment and safety** of large language models.

## Research Experience

**Agentic Reinforcement Learning for Long Horizon Task Solving** — Aug. 2026 - Present
*Carnegie Mellon University with Prof. Niloofar Mireshghallah*

- Training frontier LLM agent with RL to optimize its general abilities on **long-horizon** science benchmarks, including (self-iterated) **harness** development and **reward design**.

**Reasoning and Safety Limits of Loop Transformers** — Aug. 2026 - Present
*Carnegie Mellon University with Prof. Niloofar Mireshghallah*

- Evaluating **reasoning performance** and **safety behavior** in loop transformers to identify limitations and failure modes.

**Multi-objective Alignment with Reinforcement Learning** — Aug. 2026 - Present
*Carnegie Mellon University with Prof. Niloofar Mireshghallah*

- Developing methods to improve LLM's **safe behavior** while maintaining comparable reasoning performance.

**Reinforcement Learning for Parametric Knowledge Access in LLMs** — Jun. 2025 - Apr. 2026
*Carnegie Mellon University with Prof. Niloofar Mireshghallah*

- Discovered that RL can learn a **navigation policy** to access previously inaccessible parametric knowledge, supported by GRPO training on Qwen3 models and layerwise representation analysis.
- Conducted controlled training ablations across six SFT, rejection-sampling, and GRPO variants, identifying **on-policy sampling** as the main driver of improved factual recall.

**Drug Discovery with Generative Models and Active Learning** — May 2024 - Aug. 2025
*Simon Fraser University with Prof. Martin Ester*

- Introduced **BALD-GFlowNet**, a generative active learning framework that overcomes the **scalability limitations** of pool-based methods, achieving a **2.5-fold runtime reduction** on large-scale virtual screening with performance comparable to the BALD baseline.

## Publications

**Zhang, R.**, Kaniselvan, M., Schaeffer, R., and Mireshghallah, N. (2026). "Reinforcement Learning Improves Traversal of Parametric Knowledge in LLMs". In: *Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Oral Presentation.

**Zhang, R.** and Mireshghallah, N. (2026). "Reinforcement Learning on Benign Facts Amplifies Leakage of Memorized Private Data". In: *NeurIPS 2026 Workshop on Trust-AI-Eval (TAE)*.

Han, K., **Zhang, R.**, Wei, K., Mahdavi, H., Mireshghallah, N., and Barati Farimani, A. (2026). "SMDD-Bench: Can LLMs Solve Real-World Small Molecule Drug Design Tasks?" In: *ICML 2026 GenBio Workshop*.

**Zhang, R.***, Pandey, M.*, Cherkasov, A., and Ester, M. (2025). "Why Pool When You Can Flow? Active Learning with GFlowNets". In: *NeurIPS 2025 Workshop on AI Virtual Cells and Instruments: A New Era in Drug Discovery and Development*.

Pandey, M, Sajed, T, Semenov, I, **Zhang, R**, Ban, F, Manskaia, E, Ester, M, and Cherkasov, A (2025). "Deep Docking, Part 2: An Amplified DDU Platform for Ultra-Large Virtual Screening". In: *Under Review at ACS Central Science*.

## Awards

Dean's Honour Roll (2024), President's Honour Roll (2024), Undergraduate Student Research Award (2024) – SFU